Project Proposal

Group: **Chartreuse**

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Problem Statement

1 Business Problem

Banca Massiccia (BM) lends to businesses and loses money when borrowers fail to repay. Its loan officers currently set approval decisions, interest rates and underwriting fees using a mix of human judgment and statistical models. Because those inputs depend on how likely a borrower is to default, weak default predictions lead to mispriced loans and avoidable credit losses. BM therefore wants a more accurate estimate of each prospective borrower's one-year probability of default (PD), so that it can price loans according to risk, approve fewer loans that are likely to default, and reduce losses from bad loans without turning away creditworthy firms.

We translate this into a supervised data science problem: estimate the probability that the firm defaults within the next 12 months. Each firm-year is one observation, and the target is whether a default occurs within 12 months of the statement date. The model outputs one PD per borrower, which loan officers use for approval and risk-based pricing decisions.

1.1 The Use case Scenario

The model is used at loan origination. When a firm applies for a loan, the loan officer enters its latest annual financial statement into the model, which returns one PD over the next 12 months. This PD supports two downstream decisions:

- **Credit approval:** The loan officer uses the PD alongside other financial information and the scores of any other models the bank runs, to judge whether the borrower is likely to default in the next year. Because officers must justify and sometimes override these decisions, hence, explainability of a model is very importance to loan officers. We will use SHapley Additive exPlanations (SHAP) [5] to show which financial characteristics, such as leverage, profitability or liquidity, push an individual firm's PD up or down, and to check that the model's overall behavior is financially sensible.

- **Risk-based pricing:** The loan officer will also use the PD to set the interest rate and underwriting fees so that the loan is profitable given its default risk. A higher PD means a higher rate or fee, or a decision not to lend.

1.2 Data Science View of the Business Problem

We frame this as a supervised binary classification problem. Each observation is one firm-year, described by the firm's annual financial statement and descriptive data such as legal structure and industry. The target is 1 if the firm defaults within 12 months after the statement date and 0 otherwise. The model outputs a score between 0 and 1 that we interpret as the one-year PD. We chose this setup because of how the bank actually uses the number. A loan officer need to know how likely a firm might default rather than just a label or default. PD estimates provides the exact estimates.
But how does this solve the business problem? BM wants to lose less on bad loans. If our model works, the firms that go on to default (the bads) will consistently get higher PDs than the firms that don't (the goods). That lets the bank decline, reprice or tighten terms for the riskiest applicants. Where to set the PD cutoff is the bank's call: it depends on how much risk it can accept, and how many good borrowers it's willing to turn away to catch a given share of the bads.

Our job as data scientists is to separate the goods from the bads as cleanly as possible. We'll measure that with the Kolmogorov-Smirnov (KS) [6] test statistic, where higher is better, and with capture rate: if we flag a% of firms, we want to catch b% of the bads, with a as low as possible and b as high as possible. The PD also sets the interest rate, so ranking firms correctly isn't enough. We'll also check calibration using the Brier score [2], log-loss and calibration curves, because accurate rankings with inaccurate probabilities would still mean mispriced loans.

Solving Problem Using Data Science

2 Modeling Framework

We will build the model in stages, and we expect to loop back between them as we learn from the data. We start with the data itself, work toward features that carry a financial meaning, and then pick and test models. We judge the result on how it would perform on future borrowers, not on the data it was trained on. Following are the steps involved in our modeling framework:

- **Exploratory Data Analysis & cleaning:** Here we look at how we treat zeros, missing values. A missing value might have some predictive power so dropping it always does not makes sense.
- **Feature engineering:** Convert raw euro amounts into ratios (profitability, leverage, liquidity, coverage, cash flow, activity), crossing income-statement and balance-sheet items.
- **Feature selection:** Pick the best set of features before training any model by removing co-linearity and other feature dependencies Test by category and individually.
- **Modeling experiments:** Start with regularized logistic regression as the baseline, then compare with other complex models like tree-based models, etc.
- **Evaluation:** Fix the metrics and time boundaries up front. Use Area Under Curve (AUC) [**smith2026**], KS and capture rate for separation, and Brier score, log-loss and calibration for probability accuracy.

2.1 Exploratory Data Analysis on Sample data

We reviewed a 1,000-row sample of the training data. It covers 977 firms with annual statements from 2007 to 2012, and each row is one firm-year. A few findings shape our approach:

- **Defaults are rare, and the label horizon matters:** 93 of the 1,000 firm-years (9.3%) belong to a firm that eventually defaults, but only 21 of the 779 labeled firm-years (2.7%) default within 12 months of the statement date. Remaining defaults eventually, so a label that only checks whether a default date exists measures lifetime default, not the one-year PD. We therefore define the target with the 12-month window.
- **Missing values:** The group-equity field is entirely empty, and days receivable is missing for 69% of rows. Several debt and payables fields are missing for about 10–12% of rows, with no clear link to default.
- **The data is heavily skewed.** Total assets range from about 1.5M euros to several billion euros, with a skew of 30. This motivates ratios, logs and winsorizing.
- **Loss-making and over-indebted firms stand out.** 6.6% of the total firms have negative equity, with a default rate of 33%. Firms with negative equity eventually default.

Statement year	Firms	Eventually default (%)	Default within 12 months
2007	241	10.0	0
2008	177	18.1	3
2009	111	25.2	10
2010	116	5.2	5
2011	134	2.2	3
2012	221	0.0	not observable

Table 2: Defaults by statement year in the sample.

2.2 Feature Engineering

The data contains roughly 35 raw financial statement variables in euros, which mostly reflect firm size. These variables are highly skewed. For example, the "total asset" has a skew of 29.6 and "COGS" has a skew of 31.36. Extreme values like these dominate the fit of the logistic regression baseline and make coefficients unstable. We therefore plan to do data transformations. Applying log1p transformation to "total asset" and "COGS" reduces their skew to 1.48 and -0.73 respectively. Figure 1 shows the log1p transformation of Asset total variable.

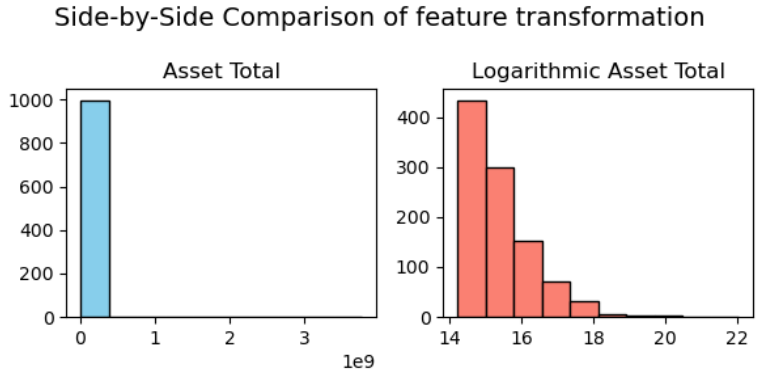


Figure 1: Asset total feature and it's logarithmic transformed compared side by side

We also plan to construct two types of ratio based features:

- *Within-category*: divide two items from the same statement to make relationships inside it explicit. Examples are operating margin (`prof_operations/rev_operating`) and the current ratio (`asst_current/debt_st`).
- *Cross-category*: divide items from different statements, such as the income statement and balance sheet, to capture relationships no single field conveys. Examples are ROA (`profit/asst_tot`) and debt-to-equity (`debt_st` + `debt_lt`) / `eqty_tot`

2.3 Feature Selection

Currently, the sample size is too small to perform any feature selection and judge the quality of the feature. Hence, we plan to run a series of steps once we receive the full dataset. We plan to run feature selection only on the training data split, so that no information from validation or later periods leaks in. We plan the following steps:

1. *Descriptive statistics*: check NULL rate and deviation for each feature, and drop those that are mostly empty $\geq x\%$ or nearly constant $\sigma \to 0$.
2. *Univariate analysis*: bin each feature and compute Weight of Evidence and Information Value (IV) against the 12-month default flag, keeping features with meaningful IV.

$$\text{IV} = \sum(\%\text{goods} - \%\text{bads}) \cdot \ln\left(\frac{\%\text{goods}}{\%\text{bads}}\right)$$

3. *Stability*: compute the Characteristic Stability Index (CSI) [8] between earlier and later statement years, and drop or re-engineer features whose distribution drifts strongly.

$$\text{CSI} = \sum(\%\text{Actual} - \%\text{Expected}) \cdot \ln\left(\frac{\%\text{Actual}}{\%\text{Expected}}\right)$$

4. *Multicollinearity*: compute pairwise correlations and within each highly correlated group, keep the feature with the highest IV and clearest financial interpretation.

2.4 Modeling Experiments

We will experiment and compare models in increasing complexity, so we can visualize what each complexity addition adds up in accuracy and what it costs in interpretability:

- *Baseline*: logistic regression with L1/L2 regularization on the selected features. It is transparent, its coefficients are easy to explain.
- *Tree-based models*: random forest and gradient boosting (e.g., XGBoost [3]. These capture nonlinearities and interactions between ratios without manual specification, and they handle missing values natively. We also plan to use other variants of boosting algorithm particularly CatBoost [7], because our data has categorical features and CatBoost can handle these kind of features.
- *Neural network*: We also plan to experiment with neural network based models, however it will be difficult to explain the predictions.

Defaults are rare, so the data is highly imbalanced. We will use class weights so that the models do not simply favor the majority class. Weighting can improve the ranking of firms but distorts the predicted probabilities, which matters because the PD sets the interest rate. We will therefore recalibrate the final scores after modeling. We also plan to tune hyperparameters with Optuna [1], which starts with random trials and then uses the results to concentrate its search on the regions of the hyperparameter space where the validation metric improves.

2.5 Evaluation

Models are evaluated at two stages: (1) during development and (2) after deployment in production. We will discuss both, but in our project we will be using development metrics. Development evaluation also has two parts:

1. *Model metrics*: statistical measures of how well the model separates defaulters from non-defaulters and how good is the model. Here we use AUC, KS, precision and recall. We will also use Receiver Operating Characteristic (ROC)[4] because it provide loan officers a full picture of the trade-off between catching bad loans and turning away creditworthy firms, so the they can choose a cutoff based on risk appetite.
2. *Business metrics*: measures of what the model is worth to the bank at a given cutoff. The `approval rate` (share of accepted applicants), the `bad rate` (share of approved loans that defaulted), and the `expected loss` (PD $\times$ LGD $\times$ EAD) summed over the approved portfolio, under assumed loss-given-default and exposure values. These are the quantities a bank tracks during underwriting, and we also plan to compare our models on these metrics.

Once the model is deployed in production, we monitor model stability. We track drift at a regular frequency (e.g., monthly or quarterly, since firms submit annual statements). We compute the Population Stability Index (PSI) of the model scores, and the CSI of key features, against the training period. If PSI exceeds a threshold (e.g., 0.25), the model is investigated and refreshed, since a distribution shift can degrade performance.

For this project we plan to hold out two evaluation sets that are never used for training or tuning: an out-of-sample (OOS) set of firms not seen in training, drawn from the same period, and an out-of-time (OOT) set of the latest statement years. OOS will test how better a model generalized to new borrowers, while OOT will test how the model performs on future periods, so we will report model and business metrics on both. We will also compute the PSI of the model scores on the OOT set to check stability.

2.6 Expected Outcomes and Next Steps

We expect our pipeline to deliver well-calibrated PD estimates that separate defaulters from non-defaulters better than a regularized logistic baseline, with SHAP explanations that loan officers can use to justify decisions. Once we receive the full dataset, we will:

1. clean the data and build the training, validation, OOS and OOT datasets
2. perform feature engineering and feature selection on the training data
3. train and tune the models, iterating on validation performance
4. evaluate the final model once on the OOS and OOT sets using the model and business metrics

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